

WOOJOO NA

• 360 Huntington Avenue, Boston, MA 02115

◇ <https://itsmeuniverse.github.io> □ na.w@northeastern.edu

EDUCATION

Northeastern University Sep. 2024 – May 2028 (expected)

Ph.D. in Electrical and Computer Engineering | Advisor: Jennifer Dy

· Research Topic: Mathematical foundations of machine learning, optimal transport, and geometrically structured learning algorithms

Tufts University Sep. 2021 – Dec. 2023

· M.S. in Computer Science, awarded Dec. 2023 (entered in the Ph.D. program)

· Master's Thesis Advisor: Abiy Tasissa

· Master's Thesis: Novel Ensemble Learning Method with Applications in Weakly Supervised Learning

University of Oxford Oct. 2016 – July 2019

B.A. (Hons) in Mathematics

RESEARCH INTERESTS

Mathematical foundations of machine learning; optimal transport on covariance and other structured spaces; monotone (descent-guaranteed) learning updates and their optimization; learning from incomplete or noisy supervision; learning theory.

PUBLICATIONS & PREPRINTS

Woojoo Na, Jennifer Dy. "ITSPACE: Monotone Gaussian Optimal Transport Updates."

International Conference on Machine Learning (ICML), 2026. Accepted as a regular paper. OpenReview · arXiv

- A proximal majorization–minimization method for the exact Bures–Wasserstein (Gaussian W_2) objective on covariances, with closed-form updates in a square-root factorization; each iteration satisfies a sufficient-decrease inequality in exact arithmetic, with a certificate-gap bound under inexact polar computations.

- First author and sole student author; led problem formulation, theoretical development, algorithm design, experiments, and writing.

Woojoo Na, Abiy Tasissa. "RACH-Space: Reconstructing Adaptive Convex Hull Space with Applications in Weak Supervision."

arXiv preprint, 2023. arXiv

- First author and sole student author; introduced an adaptive convex-hull reconstruction method for weak supervision.

- Reported state-of-the-art performance on the weak-supervision benchmark tasks considered at the time.

RESEARCH EXPERIENCE

Northeastern University Sep. 2024 – Present

Graduate Researcher, Advisor: Professor Jennifer Dy

- Developed ITSPACE, a proximal majorization–minimization method for Gaussian (Bures–Wasserstein) optimal transport between covariances: closed-form updates in a square-root factorization that preserve positive semidefiniteness and support rank-restricted factors (ICML 2026).

- Proved a per-iteration sufficient-decrease guarantee for the BW objective in exact arithmetic and a certificate-gap bound under inexact polar computations; on real-world covariance-alignment benchmarks, ITSPACE reaches low-BW-gap solutions substantially faster than BW gradient descent and entropically regularized sample-OT baselines.

- Led the project end-to-end (problem formulation, theory, algorithm design, experiments, manuscript).

Current work: monotone transport-based learning updates and computationally light, mathematically principled adaptation algorithms.

Data Intensive Studies Center, Tufts University

Sep. 2021 – Dec. 2023

Graduate Researcher, Advisor: Professor Abiy Tasissa

- Developed RACH-Space, an adaptive convex-hull reconstruction algorithm for weak supervision under partial and noisy label information.
- Built the theoretical motivation, implemented the method, and evaluated performance against weak-supervision baselines; reported state-of-the-art results on the benchmark tasks considered at the time.
- Produced a master's thesis and arXiv preprint on weakly supervised learning and ensemble methods.

University of Oxford, Mathematical Institute

July 2019 – Dec. 2019

Undergraduate Researcher, Advisor: Professor Christophe Petit

- Developed a group-theoretic trapdoor attack against Cayley hash function parameters proposed at the NutMiC 2019 conference.
- Presented the work at the 17th IMA International Conference on Cryptography and Coding. [Conference link](#)

University of Oxford, Department of Computer Science

July 2019 – Dec. 2019

Undergraduate Researcher, Advisor: Professor Thomas Lukasiewicz

- Studied similarities between transformer encoder word embeddings and conceptual-space geometry.
- Visualized embedding structure to investigate low-dimensional features shared by words in the same semantic category.

University of Oxford, Mathematical Institute

June 2018 – Oct. 2018

Undergraduate Researcher, Advisor: Dr. Andrey Kormilitzin

- Competed in the n2c2 challenge on adverse drug events and medication extraction from electronic health records.
- Implemented CNN and character-level word-embedding models for concept extraction and relation extraction from clinical text.

University of Oxford, Department of Computer Science

June 2018 – Oct. 2018

Undergraduate Researcher, Advisor: Professor Thomas Lukasiewicz

- Worked on knowledge graph completion for ontology reasoning.

SELECTED TALKS & PRESENTATIONS

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| “ITSPACE: Monotone Gaussian Optimal Transport Updates”
Poster presentation, International Conference on Machine Learning (ICML 2026), Seoul, Korea | July 2026 |
| “Novel Ensemble Learning Method with Applications in Weakly Supervised Learning”
Master’s Thesis Defense, Tufts University | Dec. 2023 |
| “Bounding Worst-Case Errors of Weak Supervision Algorithms”
Tufts University Mathematics Graduate Seminar | April 2022 |
| “Security of Cayley Hash Functions”
Invited Speaker, Ewha Womans University Cryptography and Mathematics Seminar | Jan. 2020 |
| “Trapdoor Attack on Cayley Hash Function Parameters Proposed at NutMiC 2019 Conference”
17 th IMA International Conference on Cryptography and Coding Conference link | Dec. 2019 |
| “A Deep Learning Approach to Concept Extraction from Medical Texts”
Poster Session, University of Oxford Mathematical Institute | Oct. 2018 |

HONORS & AWARDS

- Tufts University Linda M. Abriola Graduate Fellowship (\$5,000; awarded to 2 graduate students) 2022–2023
- Tufts University Linda M. Abriola Graduate Fellowship (\$5,000; awarded to 2 graduate students) 2021–2022

- Brigade Commander Award, Republic of Korea Army Start-up Competition 2021
- Lady Margaret Hall Academic Development Fund for Undergraduate Research (£500) 2018

TEACHING EXPERIENCE

Tufts University, Department of Computer Science Jan. 2023 – May 2023
Teaching Assistant, CS170 Computation Theory

ADDITIONAL EXPERIENCE

Republic of Korea Army, 2nd Artillery Brigade Feb. 2020 – Aug. 2021
Sergeant, Fire Direction Operator
- Completed mandatory military service; received three brigade-level awards for excellence and was selected for combined command-post training with the U.S. Army and Republic of Korea Army.

Utpal Shanghvi Global School, Mumbai, India July 2017 – Aug. 2017
Project Intern
- Worked on economic sustainability and public-health projects for villagers in Wada, Maharashtra, India.

LEADERSHIP & SERVICE

Oxford Invariants (Oxford Mathematics Society) April 2018 – April 2019
Treasurer

Oxford Mathematical Institute, Problem Solving Matters Program June 2018 – Oct. 2018
Mentor for Underprivileged High School Students

St. Nicholas Primary School, Oxford, UK Oct. 2016 – April 2018
Volunteer Mathematics Teacher for Immigrant Primary Students

TECHNICAL SKILLS & MATHEMATICAL BACKGROUND

Programming	Python, C++, Lean
Mathematics	Optimal transport, optimization, probability, linear algebra, topology, group theory, algebraic geometry